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Review

Light Soaking Effects in Perovskite Solar Cells: Mechanism, Impacts, and Elimination

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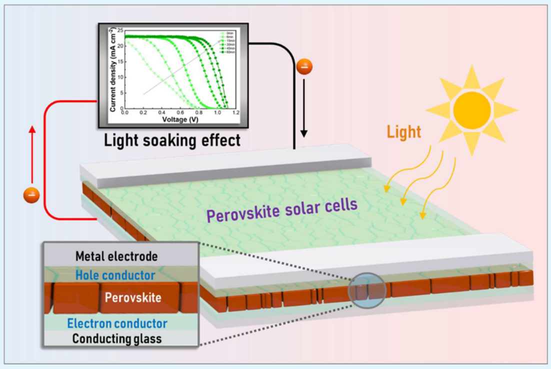
ABSTRACT: Perovskites solar cells (PSCs) have been recognized as one of the most prospective photovoltaic technologies for their combined properties of simple fabrication process, low material cost, and remarkable power conversion efficiencies of over 25%. However, the instability and poor reliability of PSCs remain the major obstacles to their practical applications. Specifically, light-soaking effect (LSE), which refers to the fluctuations of photovoltaic parameters under light exposure, represents a critical factor limiting the accuracy and stability of device power output. However, great challenges still remain in understanding and modulating the LSE in PSCs. In this review, we discuss different transient behaviors associated with LSE, and summarize various physical mechanisms (such as light-induced ions migration, trap defect passivation, lattice expansion, and charge carrier accumulation) behind the LSE together with their impacts to the device performance. Moreover, we systematically review the recent advances in developing effective approaches and strategies to mitigate or eliminate the LSE in PSCs, including interfacial modification, material doping, and surface passivation. Finally, a perspective and outlook toward LSE-free PSCs are further provided. This review offers a deeper opinion of the LSE physics with further guidance on ways to optimize the photostability of PSCs.



KEYWORDS: *light-soaking effects, perovskite solar cells, mechanism, elimination, stability*

1. INTRODUCTION

Organic−inorganic hybrid perovskite semiconductors have proven to be potential alternative photovoltaic materials for their high absorption coefficient, long charge carrier lifetime, high defect tolerance, and adjustable band gap.[1−4] Rapid development progress and high promises of perovskite solar cells (PSCs) have attracted great attention in the scientific and industrial communities. After tremendous efforts made in the composition engineering,[5−7] interface passivation,[8,9] and material optimization,[10] the power conversion efficiencies (PCEs) of single-junction PSCs have developed over 25% in just the past decade.[11] Despite their superior photovoltaic performance and cost efficiency, challenges remain in applying these materials in practical applications because of the severe environmental stability issues. Recent work has shown that the perovskite layer and interfacial contact materials in PSCs experience various degradation pathways under different accessories, such as moisture, oxygen, light, and temperature, which greatly threaten the long-term stability and reliability for industrial application of PSCs.[12−15]

Light-soaking effect (LSE) is often recognized as one of the major instability issues in PSCs. LSE means that the performance of a solar cell changes with time under a constant illumination. Specifically, LSE is shown in PSCs to enhance the performance of solar cells as a result of continuous light exposure for a certain period of time. It has been reported that this effect is reversible or irreversible after placing in the dark, which is generally related to the electrical and structural effects in the devices.[16,17] In addition to the PSCs, LSE also occurs in other solar cells, such as silicon solar cells,[18] copper indium gallium selenide (CIGS) solar cells,[19] CdTe solar cells,[20] dye-sensitized solar cells (DSSCs),[21] and polymer solar cells.[22] For example, the conversion efficiency of silicon heterojunction solar cells was reported to gain up to 0.3% due to light soaking.[18] Cabau et al. found that the efficiencies of DSSCs rise more than 4 times under open-circuit conditions with simulated AM 1.5 G illumination for 30 min.[23] Different mechanisms have been proposed to explain the LSE, such as the depletion of free electrons,[24] the migration of copper,[25] and the passivation effect of the interface.[26] Although considerable efforts were applied in investigating and explain-

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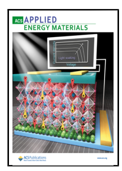

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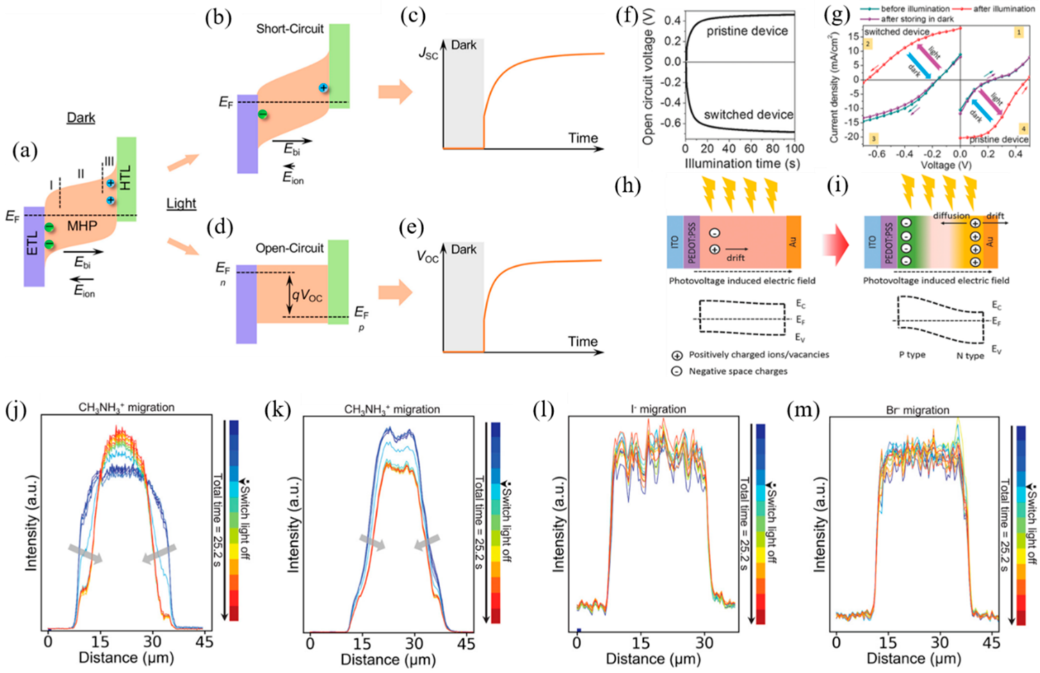

Figure 1. (a) Ion accumulation under dark resulting in energy band bending. (b) Decrease in ion accumulation under SC conditions and (d) elimination of ion accumulation under OC conditions. The rise of (c) J_{sc} and (e) V_{oc} in light. (f) Current−voltage curves of the preswitched device before illumination (blue line), after illumination (red line) for 2 min (100 mW cm^{-2}), and following placement in darkness again for 2 min (purple line). (g) V_{oc} versus illumination time for both pristine and preswitched devices. Schematic diagram demonstrating the LISP process in an OTP PV device and the energy diagram in OTP layer (h) before and (i) after LISP. The concentration distribution of the light-induced $CH_3NH_3^+$ reallocation in (j) $CH_3NH_3PbI_3$ and (k) $CH_3NH_3PbBr_3$. Concentration profiles of the light-induced (l) I^- redistribution in $CH_3NH_3PbI_3$ and (m) Br^- redistribution in $CH_3NH_3PbBr_3$. Panels a−e reproduced with permission from ref 31. Copyright 2021 American Chemical Society. Panels f−i reproduced with permission from ref 41. Copyright 2015 John Wiley & Sons, Inc. Panels j−m reproduced with permission from ref 42. Copyright 2020 Wiley-VCH GmbH.

ing the LSE in PSCs, the underlying mechanism is still under debate. Moreover, there is an explicit need to develop effective techniques to modulate and control the LSE in order to obtain reliable devices. Therefore, it is essential to review the potential mechanisms of LSE and associated strategies for elimination systematically. Here, a general overview of recent research progress on the investigations of LSE in PSCs is provided by discussing the origins and impacts of LSE in PSCs. First, we summarize different causes for the generation of LSE, including light-induced ions migration, light-induced reduced trap density, light-induced lattice expansion, and light-induced change of charge carrier accumulation. Second, we summarize the representative strategies acting on different functional layers (the electron transport layer (ETL), the perovskite absorbing layer, and the hole transport layer (HTL)) for modulating and suppressing the LSE. Finally, we discuss the future perspectives in engineering the LSE in PSCs.

2. PHYSICAL MECHANISMS OF LSE

To date, different types of the light-soaking phenomena have been observed in PSCs. Some possible origins, such as ion migration,[44] reduction of trap-state density,[51] and light-induced halide separation,[51,69] have been reported to explain LSE. Recent studies demonstrate that the LSE could be induced by the electrical and structural variation in ETL, the perovskite absorbing layer, HTL, the counter electrode, or their interfaces. The LSE in PSCs often shows large variations in the reversibility, significance, and kinetics, which depend on the device architecture, material composition, interface functionalization, and storage conditions.[27] These indicate the high complexities of the LSE and the difficulties to manage it by a versatile way. According to the previous reports, the reasons for the LSE are mostly correlated to ions migration, trap density, lattice expansion, and charge carrier accumulation. We will discuss these mechanisms in this section.

2.1. Ion Migration. Due to the ionic nature of perovskite materials, ion migration has been often observed in perovskite solar cells, which could impact optoelectronic performance and affect device operation.[28−30] Ion migration behavior could be initiated by heat, bias voltage, or light, and it has been recognized as the primary origin of LSE in PSCs. According to previous studies, Liu et al. proposed an ion redistribution mechanism to explain the LSE under short-circuit (SC) and open-circuit (OC) conditions.[31] Under SC conditions, the photocarrier transport initiated by light illumination drives the ions to diffuse and redistribute so as to reduce the interfacial accumulation, resulting in carrier separation and the increased short-circuit current density (J_{sc}). Under the OC conditions, a part of the photogenerated carriers are transferred into the ETL and the HTL under initial illumination; then the ion accumulation decreases gradually due to the lower internal built-in electrical field, leading to a completely dissipated ion accumulation and thus a gradually increased open-circuit voltage (V_{oc}) (Figure 1a−e). Huang et al. reported a fatigue behavior of $CH_3NH_3Pb_3I_3$ in PSCs by testing the devices under 12 h of continuous lighting (lights on) and then 12 h of darkness (lights off).[32] In the first 12 h of illumination, all of the photovoltaic parameters steadily increased. During the cycling tests, the device efficiency decreased after resting in the

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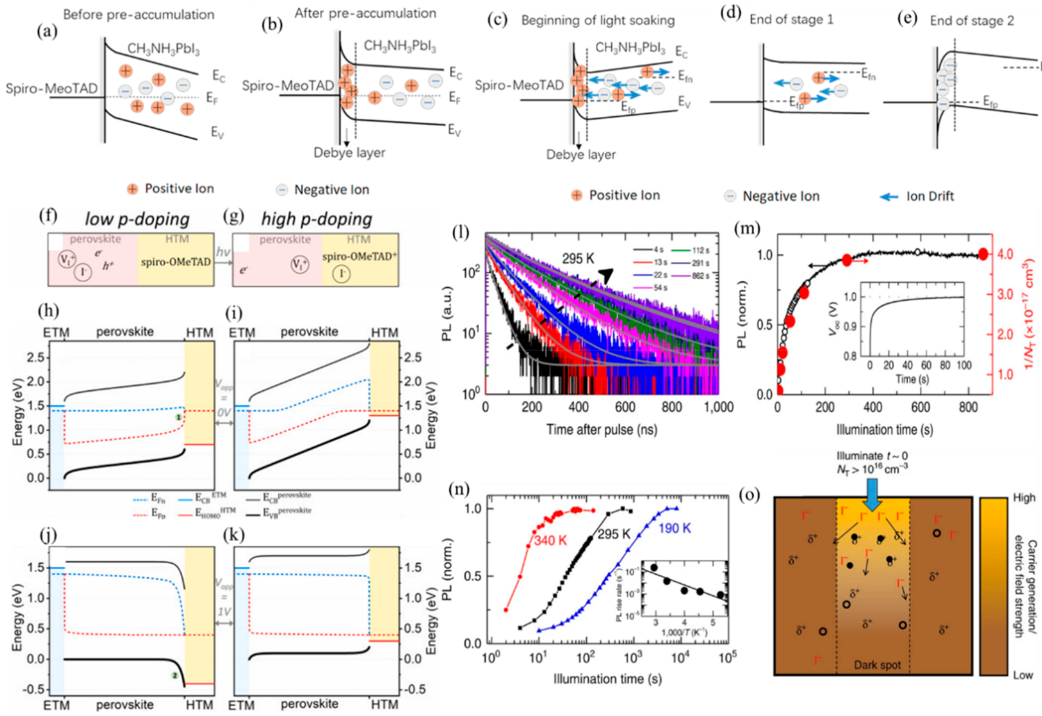

Figure 2. Energy band diagrams at the HTL/perovskite interface in the dark (a) before prestacking and (b) after prestacking. Energy band bending and ion dynamics at (c) the start of light soaking, (d) the end of the first stage of light soaking, and (e) the end of the second stage of light soaking. Diagram of the perovskite/HTM interface under light (f) at the start of the light-soaking time with the HTM having a low doping level and (g) after a long light soaking with the HTM having a high p-doping. PSC energy level diagrams showing low-p-doped (h, j) and highly p-doped HTM (i, k), with different applied voltages, V_{app} = 0 V (h, i) and V_{app}= 1 V (j, k). Number 1 in panel h indicates the large voltage drop at the perovskite interface, and no. 2 in panel j represents VB bending of the perovskite. E_{Fn} and E_{Fp} are the quasi Fermi levels of electrons and holes, respectively. (l) Series of time-resolved PL decays of a $CH_3NH_3PbI_3$ film over time under light. (m) PL over time under initial light obtained by integration of the PL decays (black open symbols) or determined from the monitored PL count rate (black solid line). Red symbols are the inverse trap densities 1/NT. (n) Normalized integrated PL over time under illumination at different temperatures. (o) Proposed mechanism of photoinduced cleaning by halide redistribution. With the lighting, electrons rapidly fill the traps, causing an electric field leading to the migration of iodide from the lighting area to fill the gap. Panels a−e reproduced with permission from ref 46. Copyright 2018 Elsevier. Panels f−k reproduced with permission from ref 17. Copyright 2020 Elsevier Ltd. Panels l−o reproduced with permission from ref 51. Copyright 2016 Springer Nature.

dark for 12 h and recovered slowly after a few hours, but the rate of recovery decreased with each cycle. Based on these phenomena, they proposed that ions diffuse in one direction during the dark conditions which seems to inhibit the performance of the device, but on the contrary it aids the performance recovery during illumination. Herterich et al. argued that the LSE is governed by ions movement in the perovskite layer.[33] They found that the observed increase in V_{oc} originates from the reduced gradient of the quasi-fermi-level (QFL) of most carriers which is caused by the redistribution of mobile ions. Xing et al. also found that the ZnO-based inverted device exhibited a light-soaking phenomenon and attributed it to the redistribution of the ions.[34]

Some work reported that light applies a significant effect on the acceleration of ion migration and ionic transport. Zhao et al. decoupled the ionic conductivity from the mixed charge transportation in order to quantitatively demonstrate light-dependent ionic transport in $CH_3NH_3PbI_3$ perovskite films.[35] The results show that light can significantly enhance ions transport by reducing the activation energies of I^-/MA^+ and H^+. Xing et al. demonstrated light-enhanced ionic conductivity in organic−inorganic halide perovskite, which reveals a reduction in activation energy for a $MAPbI_3$ film (average grain size, 300 nm) by approximately three times (from 0.27 to 0.08 eV) under a light intensity of 0.25 sun.[36] Theoretical calculation from Gottesman et al. suggests the combination of MA^+ with inorganic framework was reduced in the presence of light.[37] This light-induced softness of the lattice increases ion migration as well. Similarly, Roose and co-workers reported that a short light-soaking treatment after fabrication increases ion mobility, achieving self-passivation and the relaxation of strains, which led to improved performance for triple cation perovskites with different halide ratios.[38] Therefore, ion transfer has been widely recognized as one of the factors influencing LSE. We will then discuss different mechanisms associated with the ion migration, including cation, anions, and neutral molecules.

2.1.1. Cation Migration. Perovskites can be described by the formula ABX_3, where A is a cation, B is a metal cation, and X is a halogen anion.[39] Theoretically, these ions will move under light illumination, and therefore they apply different influences on the device performance. The A-site cation of the perovskite structure is located in the center of a cube composed of metal cations and halide anions. Recent studies demonstrate that the cation migration in the perovskite could be responsible for the LSE in PSCs.[40,41] For example, Xiao et al. demonstrated the drift of MA^+/iodine vacancies under an applied electrical field can result in a switchable photovoltaic.[40] Based on the previous electric poling and doping principles of organometal trihalide perovskites (OTPs), Deng et al. reported

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that the photovoltage-induced electric field under illumination can cause the "light-induced self-poling (LISP)" effect (Figure 1f,g).[41] They proposed that the LSE would be related to the migration of MA^+/iodine vacancies in $MAPbI_3$ once inducing photovoltage and illumination. When the device is exposed to light, the resulting photovoltage causes a destabilization of the original equilibrium state of the device established during darkness and starts to polarize the OTP layer. Consequently, the directed photovoltage electric field between the PEDOT:PSS and Au electrodes drives the drift of positive ions or vacancies in the perovskite crystal toward the Au electrode (Figure 1h). The aggregation of positive charges on the Au side increased the density of n-type doping in this region, resulting in greater energy band bending, which improved the built-in voltage and the V_{oc} of the PV devices (Figure 1i). The increase in V_{oc} propels the ions/vacancies drift further, so it forms a positive ion feedback loop. Finally, the formation of *p*-*n* doping at the anode/cathode interface will enhance the photovoltaic performance of OTP solar cells. Liu et al. developed a technique of in situ time-resolved time-of-flight secondary ion mass spectrometry (tr-ToF-SIMS) to directly observe the photoinduced ion migration.[42] When the light is on, $CH_3NH_3^+$ shows a more uniform and broader distribution between the two electrodes than the condition of under light-off conditions, indicating that $CH_3NH_3^+$ migrates from the center toward the electrodes under light-on conditions and migrates in the counter position under light-off conditions, considering the fact that there does not exist halide ions redistribution induced by light (Figure 1j−m). Li et al. also reported a strong LSE occurred in the $FA_xMA_{1-x}PbI_3$ PSC with its PCE increasing from 7.5 to 20.5% under continuous light illumination.[43] To explore the underlying process in the LSE, they used drift-diffusion simulations and performed external voltage experiments to prove ion migration was the origin of LSE. Specifically, the photovoltage generated under light illumination modulates the distribution of the mobile ions in the absorber layers, thereby increasing charge extraction capability and alleviating interfacial recombination, thus leading to the LSE in $FA_xMA_{1-x}PbI_3$(FAMA)-based PSCs. In addition, it was found that both the diffusion coefficient and the activation energy obtained from the simulations are mostly associated with the mobile cations (MA^+, FA^+), which may indicate that the mobile cations are related to the observed LSE. Zhang et al. also demonstrated that the "mobile ion" is responsible for the LSE observed in the inverted PSCs with a structure of NiO/perovskite/PCBM.[44] The authors suggest that, due to the light exposure, the mobile ions (MA^+/I^-) migrated to selective directions, forming an enhanced p−i−n junction, and the PCBM/perovskite interface can be softened and stitched by the trapped MA^+, which results in a higher V_{oc} and FF. Ye et al. reported that the light-soaking (LS) treatment has a crucial effect on optimizing the PCE of PSC with Lewis acid tris(pentafluorophenyl)borane (BCF)-doped PTAA as HTL, because it can produce more free radicals and amine cations, resulting in an enhanced hole-transport capacity in BCF-doped PTAA.[45]

2.1.2. Halide Ion Migration. Recent work also reveals that light exposure can promote the migration of halide ions to HTL, which results in a decrease of the surface nonradiative recombination and improving the conductivity of hole transport materials (HTMs). Deng et al. studied the LSE by analyzing the dynamic photoluminescence (PL) lifetime and intensity and examined its correlation with the voltage−current curves in devices.[46] They concluded that positive ions preaccumulation near the Spiro-OMeTAD/perovskite interface occurs in the devices during the fabrication. After open-circuit light immersion, the bulk electric field drifts the negative ions in the perovskite bulk to the ETL interface and minimizes the concentration of the preaccumulated positive ions (e.g. iodide vacancies) within tens of seconds, which therefore enhance the carrier separation and reduce surface recombination (Figure 2a−e). As a result, light illumination improves both the J_{sc} and V_{oc} of the devices. Tan et al. provided evidence that Spiro-OMeTAD photodoping leads to the LSE of the device performance.[17] They proposed that the photodoping of Spiro-OMeTAD under illumination at open-circuit conditions is caused by the migration of photogenerated holes and halide anions from perovskite to the formed compensated Spiro-$OMeTAD^+$ (Figure 2f−k). Due to this process, the Schottky barrier impeding the charge extraction of photogenerated holes at the interface of perovskite/Spiro-OMeTAD gradually diminishes, resulting in improved conductivity of the HTMs in the device and increased device efficiencies. Furthermore, this process is fully reversible and after storage in the dark, the poor performance of Spiro-OMeTAD is due to gradually returning to its initial neutral state. Similarly, Vu et al. suggest the photogenerated internal electric field drives I^- to drift toward the HTL side, accompanied by V_I^+ migration to the ETL site, resulting in the V_{oc} increasing from 0.72 to 0.87 V after 45 min immersion of bias light.[47] Recently, Zhang's group observed strong LSE in devices with evaporated undoped Spiro-OMeTAD as HTL. Specifically, 1 h illumination is needed in order to obtain a stabilized high efficiency of over 20%.[48] Analyzed from structural and compositional measurements, they observed the formation of oxidized Spiro-OMeTAD during the light exposure, which enables the enhancement of electrical conductivity of HTL and charge collection efficiency of devices. These results suggest the doping reaction occurs at the interface of Spiro-OMeTAD/perovskite, which mostly has its origins from the halide migration toward the Spiro-OMeTAD layer.[49]

Light can also promote the migration of halogen ions in the perovskite thin films, including the migration from an illuminated space to a nonilluminated space and from the surface to the bulk. This could effectively reduce the defect density and improve the device performance. Wu et al. reported a remarkable LS-induced enhanced performance of $CsPb(I_{1-x}Br_x)_3$ PSCs and studied the mechanism behind it.[50] They compared the LS behaviors of PSCs with different Br/I ratios and discovered that the self-healing time of device was shortened as the ratio of Br/I increased. This feature is mostly because of a higher mobility and a lower activation energy of Br^- anions. To be specific, the barrier activation to ionic migration in perovskite of Br^- (0.23 eV) is lower than that of I^- (0.29−0.30 eV). They observed that LS could accelerate halide ions migration, which result in significant defect passivation and performance improvement under LS. DeQuilettes et al. discovered that the PL lifetime and intensity of $CH_3NH_3PbI_3$ film rise significantly over illumination time (Figure 2l−o).[51] Detailed characterization shows that the rise in PL was attributed to the redistribution of iodine in the film. They proposed a mechanism that the photoexcited electrons could fill the traps and induce an electric field that causes iodide migration and trap annihilation. The I^- ions migrate from illuminated space to fill vacancies, and therefore a lower trap density and iodine concentration are generated in the light

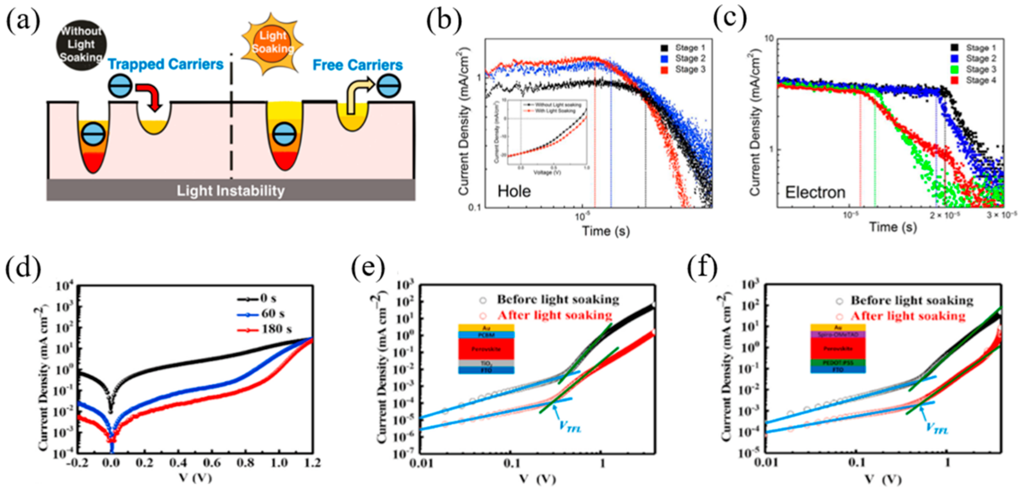

Figure 3. (a) Effect of light soaking on charge-transfer process in perovskite solar cells. (b) Temporal dependences of $CH_3NH_3PbI_3/PC_{61}BM$ bilayer cells on light illumination. (c) Electron $j-t$ profiles at 2 V bias with increasing light time. (d) $J-V$ curves measured under dark conditions after different light times. $J-V$ curves of (e) electron-only and (f) hole-only devices measured before and after light soaking under dark conditions. Panels a−c reproduced with permission from ref 59. Copyright 2016 American Chemical Society. Panels d−f reproduced with permission from ref 60. Copyright 2020 Elsevier B.V.

region. Choi et al. used temporally, spatially, and spectrally resolved optical microscopy to study the ion diffusion process from the perovskite interface to the bulk.[52] The results show that higher oxygen content together with the migration of iodine ions driven by light cures the initially defective crystal structure, and thus it enhances the internal quantum efficiency. Similar device behavior has been observed in the mesoporous carbon-based PSCs with 5-aminovaleric acid iodide (AVA) as additive. The observed slow rise in V_{oc} under illumination was ascribed to the iodide ion migration via vacancies.[53] Hoke et al. discovered the large evolution of PL intensity during LS, which was caused by higher carrier traps and radiative electron−hole recombination.[29]

2.2. Trap Density. Defects in the perovskite impact the performance of PSCs significantly, as well as on the LSE. The defects could be generated during either the perovskite synthesis or the solar cell operation.[54,55] The appearance of the defects may take place on the grain boundaries, interfaces, and the bulk of perovskites. For example, $MAPbI_3$ has shown point defects, like vacancies, interstitial, and antisite substitutional defects.[56] Meanwhile, defects are considered to be the main origin of nonradiative recombination centers.[57,58] Hence, the reduced trap-assisted charge accumulation during light illumination has been considered as the main cause of LSE.

The reduced trap-assisted charge accumulation may be originated from the decreased trap density by passivation from photogenerated electrons or photoinduced reaction with oxygen. Peng et al. studied the mechanisms behind the LSE in $CH_3NH_3Pb_3I_3$-based PSCs by combining time-of-flight (TOF) linearly increasing voltage (CELIV) measurements.[59] As illustrated in Figure 3a, upon continuous light illumination, the effect of shallow traps on current density was reduced and a reversible shallow trap filling process during LS is first directly observed in TOF measurement. Therefore, the authors proposed that LS plays an important role in reducing trap-assisted charge recombination, which increases both hole and electron mobilities and thus remarkably improves device performance (Figure 3b,c). Cai et al. reported a reversible LSE in the $CsPbI_3$-based PSCs that exhibited an increase in PCE from the previous 10.8% to 15.2% and 18.3% after 60 and 180 s soaking under AM 1.5G sunlight, respectively.[60] According to the dark current curves (Figure 3d), an increase in the built-in electric field from about 0.73 to 1.00 V after 180 s LS causes the reduced dark current and thus a higher V_{oc}. Through space charge-limited current (SCLS) measurements, they also observed decreased hole and electron trap densities in the perovskite after LS (Figure 3e,f) and proposed that passivation by photogenerated electrons leads to a lower density of defects within the perovskite, causing a reduction in nonradiative complexes associated with traps, which is beneficial to improve FF. Chen et al. revealed the influence of light intensities on the performance of $CH_3NH_3PbI_3$-based perovskite solar cells fabricated by different deposition techniques.[61] They suggest that PL efficiency is the result of competition between the defect trapping and the free electron−hole recombination under low excitation. They found that the PL increases at low-intensity illumination on a spin coating (SC) sample, suggesting that the reduction in defect trapping is due to the defect curing effect whose role is to eliminate the traps on perovskite film. Tian et al. systematically investigated light-induced PL enhancement in $CH_3NH_3PbI_3$ perovskite.[62,63] It was found that PL enhancement for large and thick crystals was much slower when the defects were found both at the surface and internal of the crystals.[63] Moreover, they speculated that the initial amplitude of the PL decay, the PL lifetime, and the PL intensity during the light illumination are ascribed to the light-induced oxygen response to passivate surface and body defects.[62] Combining this analysis, they proposed a reasonable mechanism to explain these phenomena. Initially, the light illumination can deactivate the defects close to the surface of the crystal via a chemical reaction. As the PL lifetime becomes longer and carriers' diffusion length increases, electrons and holes are able to diffuse deeper, inducing the reaction region to enter the crystal. Fu et al. attributed the dramatic time and spatial-dependent enhancement in photoluminescence of $CH_3NH_3PbI_3$ films under light exposure to a combination of trap inactivation and photogenerated carrier diffusion away from the light-exposed region.[64] With the aid of PL imaging, they directly observed the trap deactivation process, which

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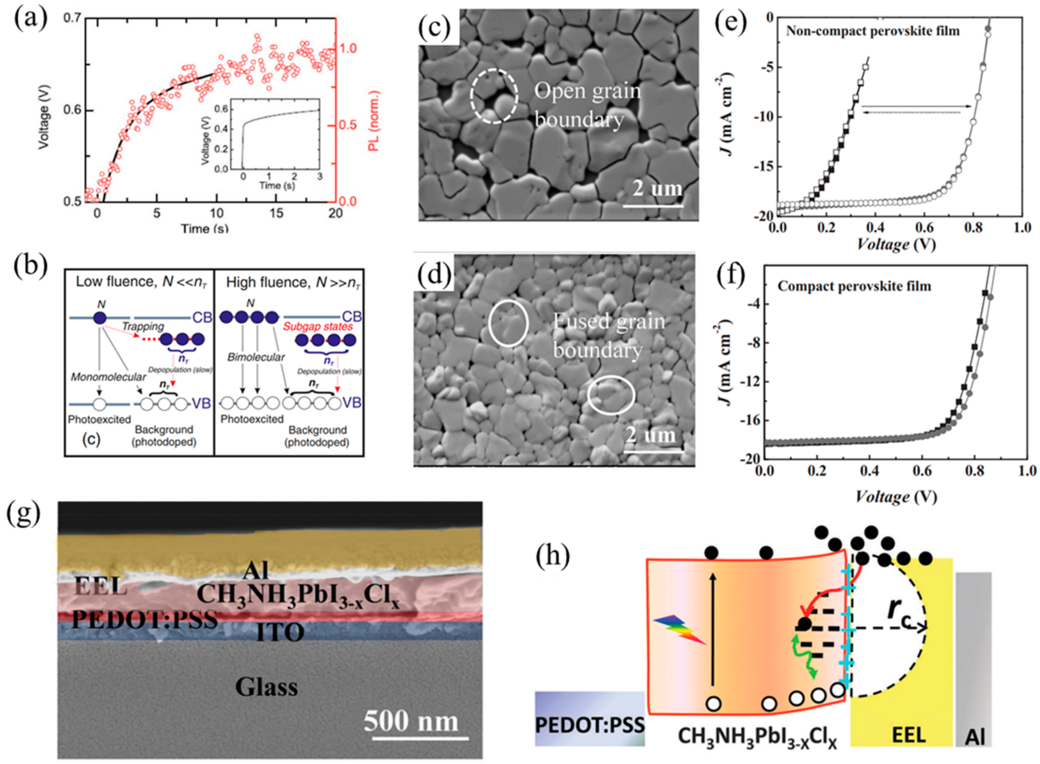

Figure 4. (a) Rise of device V_{oc} with time (black line) under light comparable to the full light intensity of the white LED light source, matching the rise in PL with time (red data) for flat film light photoexcited comparable to the full light intensity. (b) Schematic diagram describing the recombination mechanism for the low- and high-flux states. SEM morphology images of (c) nondense perovskite films and (d) dense perovskite films. $J-V$ curves of HPSCs for (e) nondense and (f) dense perovskite films. (g) SEM images of cross-sections of device structures. (h) Schematic diagram for the suggested mechanism of the light soaking. Panels a and b reproduced with permission from ref 70. Copyright 2016 American Physical Society. Panels c−f reproduced with permission from ref 76. Copyright 2016 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. Panels g and h reproduced with permission from ref 77. Copyright 2016 Royal Society of Chemistry.

results in lateral carrier diffusion over several micrometers. During the trap deactivation process, the photocarriers may diffuse laterally into the unexposed areas and the active traps may spread toward the center of the exposed area, resulting in PL enhanced lateral extension beyond the illuminated region. They further showed that the reactivation of deactivated traps is responsible for the PL decrease after illumination is turned off.

It is observed that the annihilation of Frenkel pairs also reduces the trap density. Mosconi et al. found that photoluminescence quantum efficiency (PLQE) in solution-processed $MAPbI_3$ thin films manifests strong temperature-dependent enhancement.[65] Moreover, both the PL lifetime and total intensity increase and eventually stabilize under light illumination over time, consistent with a substantial decrease in the nonradiative decay pathways. These phenomena are related to the decrease of trap density from the initial $\sim 10^{17}$ cm^{-3} to the stable $\sim 10^{16}$ cm^{-3}. From the first-principles to establish the calculation model of defect migration under light irradiation, they proposed a model to illustrate the underlying mechanism of the increased PLQE. Light accelerated interstitial iodine migration and provided kinetic impetus for the annihilation of iodine vacancy/interstitial (V_I^+/I_i^-) Frenkel pairs. The annihilation of Frenkel defects partially achieves a more perfect crystal environment and eliminates the defects associated with captured carriers. Motti et al. proposed the relative competition mechanism of light-induced defects formation and annihilation by combining ab initio simulations with PL measurements to explain PL intensity enhancement and decrease.[66] In their study, they further reveal that the long-lived electrons at trapped I_i^+ defects may promote Frenkel defect ($I_i^-V_I^+$) annihilation. The light-induced PL intensity enhancement (PLIE) process can be schematized as $I_i^+/I_i^- \cdots V_I^+ + 1e \rightarrow I_i^0/I_i^- \cdots V_I^+ \rightarrow I_i^0 +$ prist (the pristine material). This model is consistent with the temperature-dependent integrated PL results that no PLIE was observed if the temperature is too low to prevent the migration of defects.

Moreover, the state and configuration of defects change during the LS process. Wang et al. established a defect theory for LSE in lead halide perovskites (LHPs) by coordinating the defect photochemistry, ionic migration, and carrier dynamics.[67] They studied three main defects, I_{Pb}, I_i, and V_I and found various defect configurations among them, i.e., the trimer-iodine (TI) and bridge-iodine (BI) configurations. The TI and BI configurations are related to iodine migration. The combined effects of charge-state-dependent configurations, configuration-dependent carrier lifetime, and low kinetic barriers between different configurations are proposed as the reason for the giant, reversible, and bidirectional LSE. For example, when the Fermi energy level is below the valence band maximum (VBM) + 0.41 eV in the dark, the TI configurations of I_{Pb}^{1-} and I_i^{1+} dominate and act as nonradiative recombination centers. Light illumination promotes the harmful I_{Pb}^{1-} and I_i^{1+} states to capture electrons and then transfer them to the benign BI configurations (I_{Pb}^{3-}, I_{Pb}^{2-}, I_i^0, and I_i^{1-}) with longer carrier lifetimes. This corresponds to the observed process of "self-healing under the light". Guo et al. conjectured that intermixed shallow states might be formed

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in the (FA, MA, Cs)$Pb(I_{1-x}Br_x)_3$ films, which seemed to be removed into different iodide-rich, low-band-gap domains upon LS.[68]

The reduced traps can also take place in the bulk of perovskites. Combining the time-dependent PL and frequency-dependent capacitance (C−F) measurement, it is found that LS can reduce the density of charged bulk density through the photogenerated electrons trapping process, leading to enhanced carrier transfer to the electrodes and increased FF during LS.[69] Stranks et al. discovered that the PL of $MAPbI_3$ films increased slowly on a time scale of a few seconds in the presence of light, which corresponded to the rising trend in V_{oc} of planar heterojunction halide PSC (HPSC) in simulated (air mass of 1.5 G) sunlight (Figure 4a,b).[70] These results demonstrate the same potential mechanisms to support a better understanding of PSCs. These slow rises in device performance and PL derive from the slow filling and stability of the bulk charge trap state are caused by the injection or photogeneration of electrons, which improves the QFL of the electrons and thus increases V_{oc}. The slight and slow rise time suggested that ion movement could also be the cause of the decrease in trap density with time or the stabilization of the trap charge under illumination. Galisteo-Lopez et al. found that photogenerated carriers could effectively fill traps in the bulk of perovskite film with the aid of oxygen-related species, which are responsible for the dramatic increase in PL intensity.[71] For example, Friend et al. studied the origin of improved performance of LSE in SnO_2-based PSCs and observed the downward shift of Fermi level.[72] Electrochemical impedance spectroscopy (EIS) and Mott−Schottky analysis indicated that LS caused the passivation of one or more n-type defects. Combining with X-ray photoelectron spectroscopy (XPS) data, they concluded that the existence of oxygen vacancy (V_O) and H_O was the main cause, and a lower number of V_O resulted in a reduced LSE.

The traps at both interfaces of perovskites could be reduced by light illumination. Hong et al. found that the fresh and aged devices had almost the same maximum V_{oc}, while the initial V_{oc} in the aged devices were smaller than those of the fresh devices, indicating that there was a more severe LSE in the aged device.[73] The reason for that is iodine ions diffuse from the interface of CuI and $MAPbI_3$ toward the interface with PCBM during the aging process of the device, thus reducing the internal field and initial V_{oc}. Upon LS, photogenerated charge carriers fill the trap mediated by the presence of ions and neutralize the interface defects, which will restore the built-in voltage and lead to an increased V_{oc}.

The reduced traps can also occur at the perovskite/HTL interface. Wu et al. examined the kinetics of hole transfer from $CH_3NH_3PbI_3$ perovskite to Spiro-OMeTAD by testing the PL properties of perovskite with and without Spiro-OMeTAD.[74] The PL intensity and lifetime of the perovskite without Spiro-OMeTAD increased under continuous light, while, with Spiro-OMeTAD, the quenched PL is reduced and the hole-transfer efficiency between the two layers is enhanced by ∼6 times under 30 min light exposure. They proposed two plausible explanations. One is that during light-soaking photogenerated charge carriers passivate the perovskite/Spiro-OMeTAD interfacial traps, leading to a more efficient hole-transfer process. The other is the enhanced crystallization of perovskite films and the reduced trap density during LS facilitate free holes diffusing to the perovskite/Spiro-OMeTAD interface. Zheng et al. performed steady-state PL and time-resolved PL spectroscopy to investigate the slow response of carrier dynamics on $(FAPbI_3)_{0.85}(MAPbBr_3)_{0.15}$ perovskite under continuous illumination.[75] The PL enhancement of the Spiro-OMeTAD-covered perovskite interface was more pronounced than that of the Spiro-OMeTAD-uncovered perovskite interface under continuous constant illumination. Both defects curing process and the migration of mobile ions affect the dynamic response. When perovskite is contacted with Spiro-OMeTAD, the defect-trapping ratio of carriers during the defect curing process is significantly reduced under continuous illumination and the evolution of band curvature caused by ion migration leads to a decrease in the hole-transfer rate from the perovskite interface to Spiro-OMeTAD. These carrier dynamics are positive to the radiative recombination of carriers and thus lead to PL enhancement.

The reduced traps can take place at the perovskite/ETL interface. Shao et al.'s work proposed that trap states at the interface of the perovskite dominate the LSE.[76] For non-compact perovskite films, solar cells applying PCBM as ETL show a severe LSE, with the PCE improved from 3.7% to 11.6%. However, devices using compact films with smaller grain sizes show a negligible LSE of PCE, increasing slightly from 11.4% to 11.9%. By analyzing the device parameters, PL, and IS measurements, they conclude that the trap states can be populated by photogenerated carriers and mobile ions during LS, especially the positively charged iodine vacancies (Figure 4c−f). Shao et al. argued that the trap-assisted recombination at the perovskite and ETL interface is the cause of the LSE in HPSCs (Figure 4g,h).[77] It is shown that devices using fulleropyrrolidine with a triethylene glycol monoethyl ether side chain (PTEG-1), which has a dielectric constant of 5.9 as ETL in HPSCs, showed a negligible LSE with the PCE increasing from 15.2% to 15.7%, while the device using $PC_{60}BM$, which has a dielectric constant of 3.9 as ETL, showed severe LS with the PCE improving from 3.8% to 11.7%. From PL spectroscopy and IS measurements, they proposed that the LSE of PTEG-1 is negligible, possibly due to the higher dielectric constant and electron donating properties of PTEG-1, which may reduce the trap-assisted recombination and thus eliminate the LSE in HPCs. Wang et al. considered that the ETL/perovskite interface is the major cause of the LSE in planar heterojunction PSCs.[78] Their research showed that the photogenerated electrons would drift toward the TiO_2/perovskite interface and accumulate there, generating an electric field that impedes charge transport. Under prolonged exposure to light, the defects at the interface and in TiO_2 can be neutralized and thus the charge transfer will gradually be improved, eventually enhancing the device's performance.

2.3. Lattice Expansion. Light-induced lattice expansion is also proposed to rationalize why light exposure could enhance the performance of HPCs.[79−84] Tsai et al. observed the PCE increased from 18.5% to 20.5% under a continuous 1 sun (100 mW/cm^2) light illumination in a device with mixed-cation pure-halide perovskite. They suggested that this LS behavior was related to the light-induced lattice expansion rather than the ion migration.[79] Similarly, Liu et al. found that FA cation-based MHP shows negligible photoinduced lattice expansion, whereas the other three cations (MA, Cs, or PEA) based MHPs exhibit much stronger lattice expansion under light exposure.[82] Calculations by density functional theory (DFT) further confirmed this result, revealing that A-site cation plays a key role in photoinduced strain and lattice expansion of MHPs.

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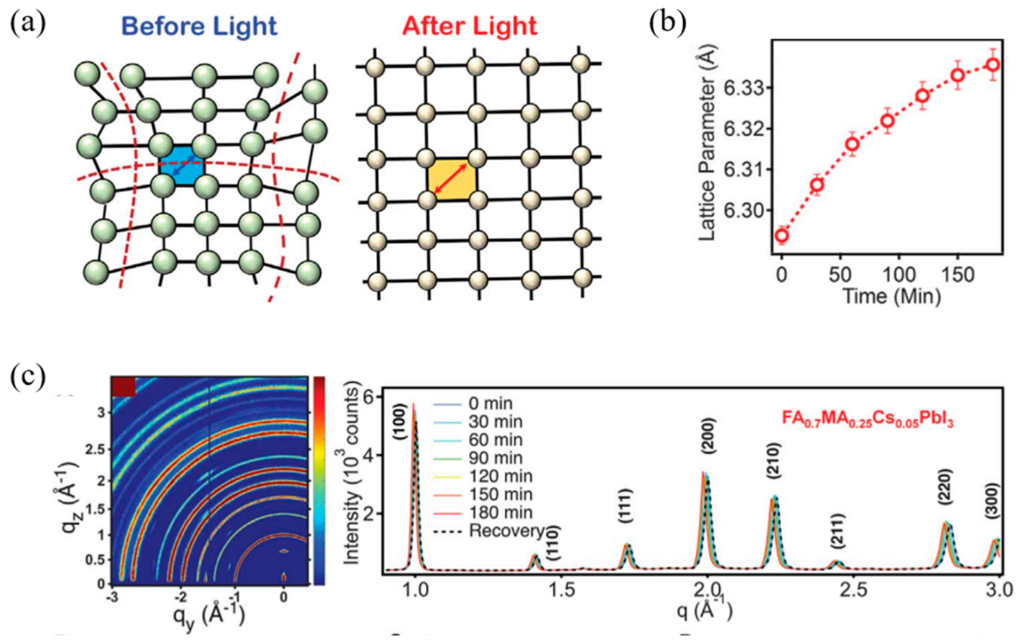

Figure 5. (a) Schematic describing the crystal structure change before illumination (local distortion) and after illumination (lattice expansion). (b) Typical synchrotron diffraction diagram (left) and line cut of GIWAXS diagram (right) for $FA_{0.7}MA_{0.25}Cs_{0.05}PbI_3$ (cubic phase) films at different light times and recovered spectra acquired after keeping the films under darkness for 30 min. (c) Variation of lattice constant with time of illumination. Panels a−c reproduced with permission from ref 79. Copyright 2018 The American Association for the Advancement of Science.

As reported, the lattice expansion could be induced by the weakened covalent bonds because the photogenerated electron−hole pairs could reduce the distortion of Pb−I−Pb bonds and make Pb−I bonds elongate under light (Figure 5a).[79] Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements showed that all diffraction peaks shifted toward lower values of scattering vector **q** with increasing illumination time (Figure 5b), corresponding to an isotropic increase in lattice constant d (Figure 5c). In situ device characterization and quantitative device modeling jointly indicate that the lattice expansion relaxes the lattice strain, thereby lowering the energy barrier at the contact interface and reducing the nonradiative combination at the perovskite crystal and interface. These effects result in the improvement of V_{oc} and FF.

Researchers also investigated the influence of lattice expansion on radiative recombination.[80,83] Wang and Long studied the impact of light-induced lattice expansion on the nonradiative charge recombination in $FA_{0.75}MA_{0.25}PbI_3$ perovskite by using the ab initio nonadiabatic molecular dynamics simulation coupled with time-dependent DFT.[83] The simulations show that the lattice only expands for 0.64% under light irradiation, which leads to a slightly increased Pb−I−Pb bond angle and Pb−I bond length, releasing more space for atomic motions. The enhanced atomic fluctuations accelerate quantum decoherence and increase electron−photon coupling. At the same time, lattice expansion introduces several high-frequency phonon modes, which further reduce the decoherence time. As a result, the decoherence time was shortened by 37% and the electron−phonon coupling was increased by 13%. In other words, rapid decoherence plays important roles in suppressing the nonradiative electron−hole recombination, thus improving the perovskite photovoltaic and photoelectronic device performance.

Kim and Hagfeldt investigated the effect of photoinduced crystal structural change on recombination behavior in perovskite materials based on trications (FA^+, MA^+, and Cs^+) and dihalides (I^- and Br^-).[81] Under continuous light irradiation, X-ray diffraction pattern gradually shifts to lower angles, which indicates a photoinduced structural change. In addition, the lattice expands in a specific direction, giving the crystal a cubic-like structure with enhanced symmetry. The improved crystal symmetry increased the antibonding overlapping, being responsible for a slightly decreased band gap energy (E_g). Therefore, the decrease in the trap density of perovskite and the inhibition of nonradiative recombination lead to a higher photocurrent of the devices.

Zhang and Wei unveiled the underlying mechanisms of how lattice expansion affects nonradiative recombination dynamics, thus enhancing the solar conversion efficiency in hybrid perovskites.[84] They demonstrated that the iodine interstitial (Ii) was the dominant nonradiative recombination center in $FAPbI_3$ and the involved lattice expansion could reduce the nonradiative capture coefficient and increase the carrier lifetime. However, the lattice expansion has no effect on the capture coefficient while it changes the band gap or leap energy level. Because the electron capture is the rate-limiting step, the leap energy level does not change. Instead, lattice expansion affects the capture barrier through lattice softening and lattice relaxations. On one hand, the softened lattice caused by lattice expansion decreases the curvature of the potential energy surface, resulting in a reduced capture barrier. On the other hand, lattice expansion leads to lattice relaxations and charge-state transitions, which increases the capture barrier. As a result, the latter effect is stronger than the former one and the electron capture barrier increases in the expanded lattice, followed by efficiency enhancement. However, there were some debates regarding thermally induced lattice expansion. By controlling the temperature under dark and illuminated conditions, Rolston et al. showed that the lattice expansion mechanism is consistent with heat-induced thermal expansion

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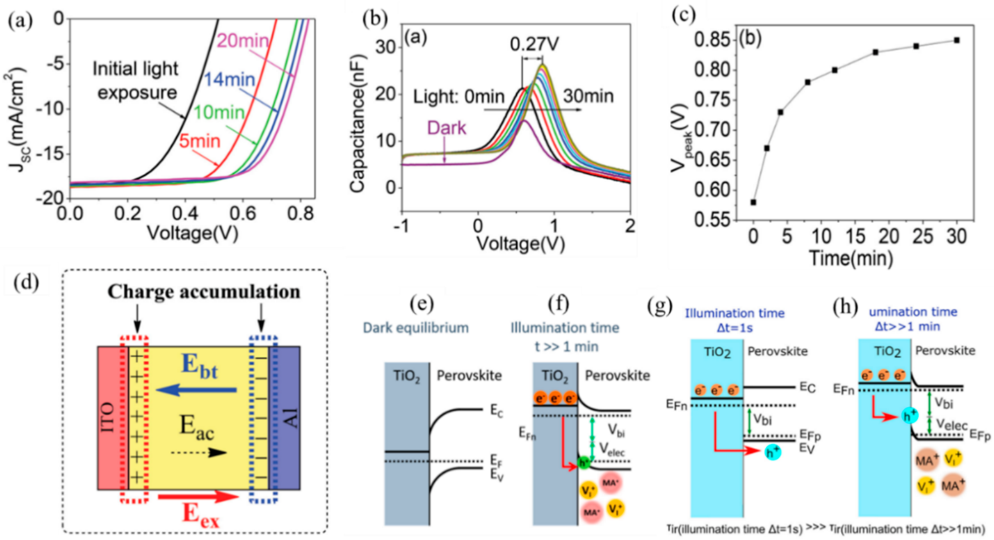

Figure 6. (a) Influence of light soaking on photovoltaic performance for ITO/PEDOT:PSS/perovskite/PCBM/Al device under light. (b) Variation of $C-V$ characteristics of ITO/PEDOT:PSS/perovskite/PCBM/Al device under light for 30 min compared to dark condition. (c) Change of V_{peak} as a function of light illumination time. (d) Schematic of the integral electric field of perovskite solar cells. Energy diagram at the TiO_2/perovskite contact under open-circuit conditions (e) in dark where there are no migrating positive cations and vacancies and (f) after a prolonged lighting time ($t \gg 1$ min). Energy diagram at the TiO_2/perovskite contact at two different stages: (g) at a very short time with only red LED pulse lighting (t = 1 s) and (h) after a large amount of light-soaking time of $t \gg 1$ min. Panels a−d reproduced with permission from ref 69. Copyright 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. Panels e and f reproduced with permission from ref 88. Copyright 2017 American Chemical Society. Panels g and h reproduced with permission from ref 89. Copyright 2016 Elsevier Inc.

during LS.[85] So the mechanism of lattice expansion associated with LSE needs further investigation.

2.4. Charge Carrier Accumulation. The LSE also originates from the change of charge carrier accumulation upon light illumination. For example, the light exposure could reduce the charge carrier accumulation at the electrode interface, leading to an increased V_{oc}. Zhao et al. found that the device with a structure of ITO/PEDOT:PSS/$CH_3NH_3Pb_{3-x}Cl_x$/PCBM/Al exhibited a reversible LSE with significant increase of V_{oc} and FF upon light exposure (Figure 6a).[69] From capacitance−voltage ($C-V$) measurements (Figure 6b,c), V_{peak} values become larger as light exposure time increases, indicating that the accumulation of photogenerated charge carriers at the electrode interfaces is less. Light exposure can decrease the charge accumulation at the electrode interface via the two following mechanisms (Figure 6d). First, the photogenerated carriers can fill the defects at the electrode interface. Second, the built-in electric field is changed by ions migration, which influences charge accumulation at the electrode interfaces. Upon LS, these two ways can increase the interfacial potential barrier at the electrode interfaces and thus increase the V_{oc} to a large extent. Srivastava et al. studied the defect density of state (DOS) during LS for chlorine-based mixed HPSCs.[86] $C-F$ and $C-V$ measurements are performed to probe the response of the defects and to characterize the reduced electrode surface accumulation of photogenerated charge carriers under light illumination. The DOS value of the defect decreased from 10^{16} to 10^{15} eV cm^{-3}, and the peak energy dropped from 0.27 to 0.26 eV, contributing to the reduced surface accumulation and the increased value of V_{oc}. Lee et al. demonstrated that 1 sun light irradiation can trigger the generation of free carriers which can neutralize and decompose accumulated trap states and charges to restore device performance after ultraviolet (UV) degradation.[87]

Others hold the view that light can increase the charge accumulation at the interfaces, inducing the occurrence of interfacial band bending and electrostatic potential, thus the enhanced V_{oc}. Hu et al. studied the photovoltage of $CH_3NH_3PbI_{3-x}Cl_x$ solar cells over different light intensities and proposed that the interface between ETL and perovskite plays a significant role in photovoltage behavior.[88] Under continuous LS, the holes are accumulated at the interface between TiO_2 and perovskite, inducing the interfacial band bending. Then an additional electrostatic potential, V_{elec}, was created, which contributed to the built-in potential, V_{bi}, and consequently increased the whole photovoltage (Figure 6e,f). The dynamic influence of LS in PSCs on photovoltage decay and the kinetics involved were studied by Gottesman et al. via open-circuit voltage decay (OCVD) spectroscopy.[89] As the LS time prolongs, the slow mobile ion charges are accumulated at the double layer contact which sustains an electrostatic potential. The electrostatic potential results in higher charge accumulation at both sides of the interface and thus the increased recombination in the contact (Figure 6g,h). Zarazua et al. concluded from their explanation of capacitance changes that the large photovoltage origins from the accumulation of electronic carriers at the contact, thus forming upward band bending at high voltage.[90] Hu et al. studied the steady-state PL of $CsPbBr_3$ films with various thicknesses under illumination on different contacts, and they proposed that under light illumination an upward band bending at the perovskite led to a higher charge accumulation at the interface, attracting photoexcited electrons toward the interface and enhancing radiative recombination.[91] Yang et al. proposed a model that combines ion migration and charges accumulation at the interface between electron selective layer and perovskite to explain the LSE.[92] As the LS time increases, the movement of ions such as MA^+ ions increases. After LS, the mobile ions will accumulate at the opposite interface due to photopolarization.

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2.5. Other Mechanisms. In addition to these four main mechanisms associated with LSE, there are also some other causes, such as the passivation effect of oxidized PbO bond networks, the formation of KBr-like compounds, and light-induced halide dealloying. For example, Godding et al. proposed a comprehensive mechanism to explain the "photo-brightening" (the increase in photoluminescence during exposure to light in an ambient atmosphere) phenomenon observed in metal halide perovskites.[93] Based on detailed characterization, they suggested that atomic lead generates with a loss of iodine at the initial light-induced step. Oxygen is physically absorbed on the surface to form superoxide, and a proton is extracted from MA^+ cation to form peroxide. The peroxide oxidizes with atomic lead mainly at the perovskite surface, yielding Pb−O bond networks which reduce the number of charge-trapping states and lead to the observed photobrightening. Zheng and co-workers compared the photostability of 0 mol % K^+ with 3.5 mol % K^+-doped mixed-cation and mixed-halide perovskite ($Cs_{0.05}(FA_{0.85}MA_{0.15})_{0.95}Pb(I_{0.85}Br_{0.15})_3$), and they found that the latter showed an improvement of photophysical properties under light illumination.[94] It was confirmed that the light-triggered passivation effect occurred in the K^+-doped perovskite, which plays a critical role in the improved PSC performance. From the scanning transmission electron microscopy (STEM) results, they found that the distribution of I was relatively homogeneous, while Br and K had a similar spatial distribution (Figure 7a−c). The authors proposed that the KBr-like compounds were formed in the K^+-doped perovskite film under light exposure. The formation of strongly bonded KBr-like compounds could reduce the interfacial trapping defect density and confine the ion migration. Besides, it facilitates the homogeneous "I^- to Br^-" halide substitution process, resulting in an improved PSC performance. It has been reported previously that the light-induced dealloying phenomenon occurred in $CsPb(Br_xI_{1-x})_3$ perovskite.[95] Niezgoda et al. also found the light-induced halide dealloying (as shown by the red shift of PL peak in Figure 7d and the shift of XRD peak in Figure 7e) and further concluded that it has a critical effect on increased efficiency (9.2 ± 0.64% on average with a champion PCE of 10.3%) of $CsPbBrI_2$-based solar cells.[96] As shown in Figure 7f, the $CsPbBrI_2$-based device has a V_{oc} of 0.93 V, J_{sc} of 1.94 mA/cm^2, and an FF of 19.45% at initial illumination. After 20 min of constant illumination, the V_{oc}, J_{sc}, and FF increased to 1.10 V, 12.19 mA/cm^2, and 68.69%, respectively. With the aid of PL and X-ray diffraction, it is suggested that light-induced dealloying of $CsPbBrI_2$ can improve the hole mobility in $CsPbBrI_2$ and induce a higher proportion of holes to Spiro-OMeTAD. In other words, the increased solar cell efficiency under light exposure originates from the improvement of hole collection in devices due to dealloying of halides. Moreover, the dealloying process turns out to be reversible; the device reverts to lower PCE once the light is removed. Liu et al. disclosed an irreversible LSE in PSCs using TiO_2 as an ETL.[16] The device efficiency was optimized after 15 min of light illumination but did not recover in the dark for 4 days. They proposed a V_O associated mechanism (Figure 7g). First, light exposure can increase the film conductivity of TiO_2, because TiO_2 is an n-type semiconductor and the V_O can be a donor in it. Second, the charge extraction process between perovskite crystallites and the ETL can be accelerated. Third, it could lead to an enlarged built-in potential of PSCs and thus the increased V_{oc}. On the basis of DFT calculations and the observed extremely slow photoconductivity response of $CH_3NH_3PbI_3$ perovskite, they proposed a light-induced MA^+ rearrangement mechanism where light illumination weakens the MA^+/inorganic frame binding, and thus the rotational freedom of the MA cations is enhanced.[37] As illustrated in the cross-sectional electron-beam-induced current (EBIC) measurements, Kedem et al. found the carrier diffusion length in $MAPbBr_3$(Cl)-based solar cells increases from Ln = 100 ± 50 nm in the dark to Ln = 360 ± 22 nm in the light conditions.[97] The results showed that the increase of the diffusion length was related to the photo-generation of carriers, but not to the absolute density of carriers. Recently, the Zhang group reported the anion modulated chemical doping process of Spiro-OMeTAD also influences the LSE of devices.[98] Specifically, the degree of delocalization (DOD) of the counterions (PF_6, TFSI, and PFSI) in lithium salts determines the cation exchange between Li^+ and Spiro-OMeTAD$^+$, which influences the strength and reversibility of LSE. For example, PF_6^- with a low DOD resulted in a strong and reversible LSE. Under light illumination, there are insufficient mobile PF_6^- to compensate spiro$^+$; thus these unpaired spiro$^+$ would recombine with electrons. As the illumination time increased, more charged spiro$^+$ produced stronger electrostatic interaction between spiro$^+$ and PF_6^-, promoting the anion exchange from $Li^+PF_6^-$ toward spiro$^+$-PF_6^-. This process could stabilize the spiro$^+$, boost the hole mobility, and reduce carrier recombination, which explain the enhancement of device performance. Moreover, the LSE in the devices was quite reversible,

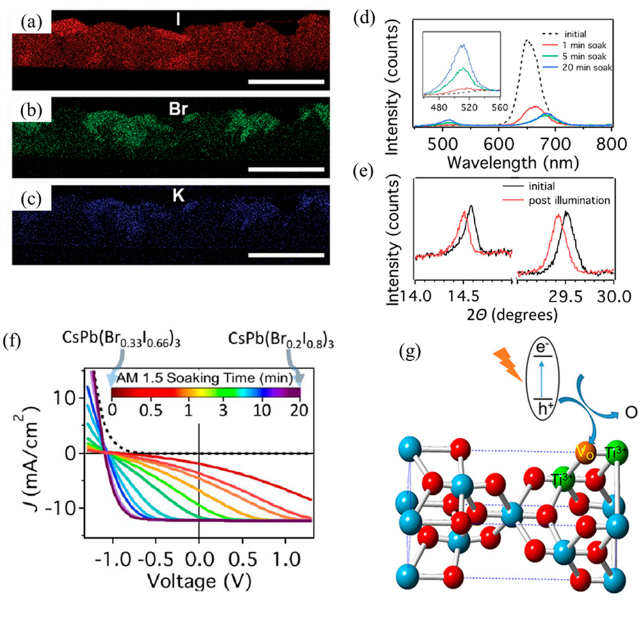

Figure 7. (a−c) STEM-EDX elemental mapping of I, Br, and K in 3.5 mol % K^+ perovskite film with all scales representing 1 μm. (d) PL spectra of the $CsPbBrI_2$ film. (e) XRD spectra of Kapton-film-sealed ITO/TiO_2/$CsPbBrI_2$/Spiro-OMeTAD devices. The transition to low 2θ after light soaking demonstrates the formation of a large number of I-rich regions in PV devices. (f) Effect of light-soaking time on the J−V curve. (g) Schematic diagram of oxygen vacancies (V_O) generation on TiO_2 surface. Panels a−c reproduced with permission from ref 94. Copyright 2019 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. Panels d−f reproduced with permission from ref 96. Copyright 2017 American Chemical Society. Panel g reproduced with permission from ref 16. Copyright 2017 AIP Publishing.

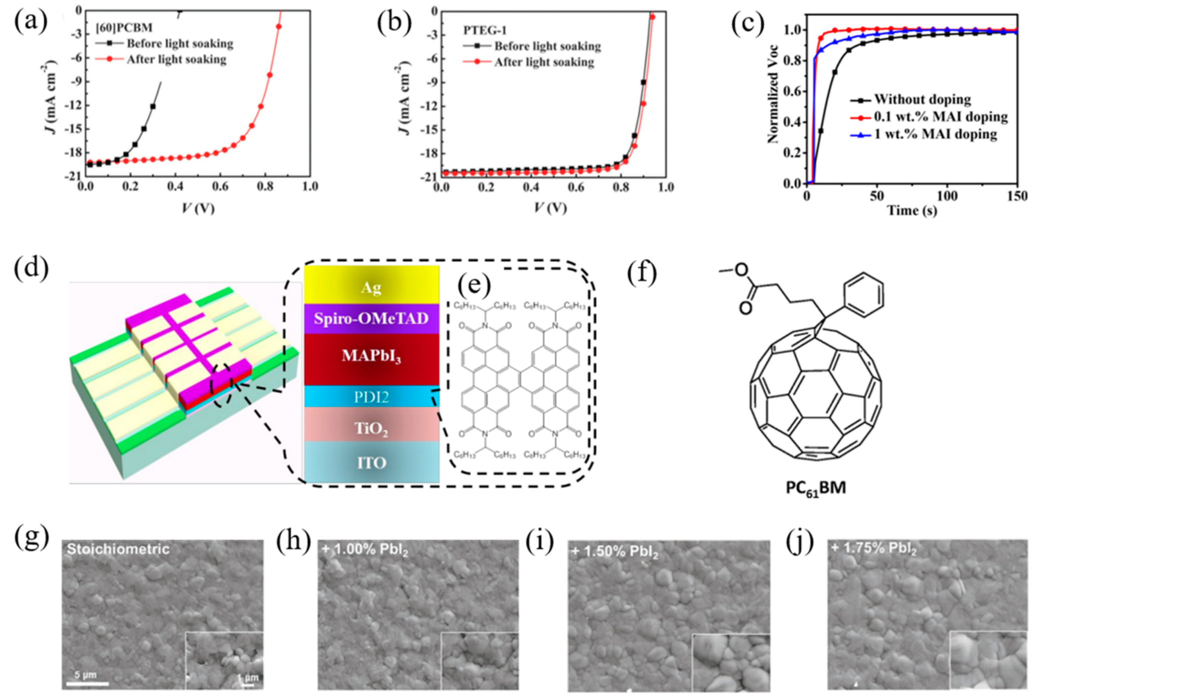

Figure 8. $J-V$ characteristics of devices under lighting using (a) $PC_{60}BM$ and (b) PTEG-1 as ETLs. (c) Evolution of V_{oc} with time in PSCs using PCBM with different MAI doping concentrations. (d) Structures of the perovskite PV device and (e) chemical structure of PDI_2. (f) Molecular structure of $PC_{61}BM$. Top-view SEM images of perovskite films prepared with (g) stoichiometric, (h) + 1.00%, (i) + 1.50%, and (j) + 1.75% PbI_2. Panels a and b reproduced with permission from ref 77. Copyright 2016 Royal Society of Chemistry. Panel c reproduced with permission from ref 44. Copyright 2017 American Chemical Society. Panels d and e reproduced with permission from ref 102. Copyright 2018 Royal Society of Chemistry. Panel f reproduced with permission from ref 78. Copyright 2018 American Chemical Society. Panels g−j reproduced with permission from ref 50. Copytight 2021 Wiley-VCH GmbH.

indicating this light-induced anion exchange between spiro$^+$-PF_6^- and Li^+-PF_6^- could be a reversible process. To prove this mechanism, an ion coordination approach was developed to modulate the electrostatic interaction between ions, which enabled suppression of LSE in devices.

3. SUPPRESSION OF THE LSE

Although LSE is beneficial for the improvement of photovoltaic performances of perovskites, it may still cause detrimental effects on the device as the output is unstable, so it is considerable importance to control and eliminate the LSE for a steady output in future real-life applications.[99,100] We will summarize and discuss the recent strategies for controlling the LSE from different layers of solar cells in the following sections.

3.1. ETL. ETL plays a critical role in receiving and transporting electronic carriers in solar cells, so the effectiveness and stability of ETL material significantly affect the efficiency of the devices. Various methods, such as structural optimization of electron transport materials, chemical doping of electron transport materials, and surface treatment of the ETL, have been developed to control the LSE. For example, Shao et al. employed fulleropyrrolidine with a triethylene glycol monoethyl ether side chain (PTEG-1) with a dielectric constant of 5.9 instead of $PC_{60}BM$ (3.9) as an ETL; the LSE can almost be eliminated, and the device efficiency has increased considerably (Figure 8a,b).[77] Comparatively, those devices with $PC_{60}BM$ as the ETL exhibits strong LSE, with a significant increase in PCE from 3.8% to 11.7%. While the device with the ETL using PTEG-1 shows negligible LSE, with the PCE minorly improving from 15.2% to 15.7%. This is due to the higher dielectric constant and electron donating property of PTEG-1, which contributes to recombination compounding between traps and free electrons and passivates the electrons traps, respectively. The decrease of the trap-assisted recombination consequently increases performance of the solar cell and reduces the LSE. Zhang et al. modified the electron-transfer material PCBM with 0.1 wt % MAI doping to prepare PSCs with fewer LSE.[44] As shown in Figure 8c, the device based on undoped PCBM took ∼5 min to reach a stabilized V_{oc} after light exposure, while the device using 0.1 wt % MAI-doped PCBM is much faster (complete in <1 min) to stabilize. The reduced light-soaking time is attributed to the supplement of trapped MA^+ in the MAI-doped PCBM, so the MA^+ does not need to travel a long distance to the PCBM interface, facilitating enhanced p−i−n junction formation. However, overdoped devices may need more time for ion migration to reach a new quasi-static state, as shown in Figure 8c. The strength of LSE became even stronger in the devices with 1 wt % MAI doping. It was also reported that when [6,6]-phenyl-C_{61}-butyric acid methyl ester ($PC_{61}BM$) was incorporated with 0.5 wt % MAI, the LSE was almost disappeared for the NiO_x devices due to the efficient charge transfer. And the device stability and performance were improved (the champion PCE is 19.05%, and the FF is 0.84).[92] Roose and Friend found Ga-doped SnO_2-based devices exhibited reduced LSE due to their fewer V_O, which consequently led to reducing nonradiative recombination.[72] Méndez et al. treated the surface of SnO_2 with UV−ozone to improve the performance of the PSC and reduce its LSE.[101] From impedance spectroscopy measurements, it was explained that the treated samples showed better wettability, increased resistance to recombina-

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tion, and decreased V_O, leading to efficient charge collection at the ETL/perovskite interface as well as outstanding properties.

3.2. Perovskite. The uncontrollable morphology and poor crystallinity of perovskite films prepared by solution processing are the main reasons that affect the device's stability. Interfacial modification can effectively passivate defects and improve film quality, which reduces nonradiative recombination and improves device performance. Yang et al. synthesized and explored a molecule perylene diimide (PDI_2) with π-conjugation between TiO_2 and perovskite in a device with a structure of ITO/TiO_2/$MAPbI_3$/Spiro-OMeTAD/Ag (Figure 8d,e), and a champion PCE of 19.84% was achieved with a shortened stabilization time from about 30 s to less than 10 s.[102] The enhanced device efficiency and the restrained LSE are a consequence of suppressed interfacial recombination from two aspects. First, the PDI_2 has high conductivity, which favors the electron transport and inhibits interfacial accumulation and recombination. Second, the coordination of oxygen atoms of carbonyl groups of PDI_2 with Pb^{2+} in the perovskite crystal contributes to the passivating trap density of the crystals and suppressing interfacial recombination. Similarly, Wang et al. introduced PDI_2 and $PC_{61}BM$ to modify the interface between TiO_2 and perovskite and as a result the LSE was decreased and dispelled.[78] Under simulated irradiation of AM 1.5 G at 100 mW/cm^2, two devices achieved PCEs of 15.7% and 17.7% respectively, compared to 12.4% for the reference device, due to the presence of passivation defects at the interface and reduced charge accumulation. The reduced LSE mainly benefited from PDI_2's superior properties, such as high conductivity, π-conjugated structure, and the blunting effect of the electron-rich oxygen element in the amide moiety. For $PC_{61}BM$, the elimination of LSE mostly originates from the superior properties of proper energy level matching, and the passivating effect due to its ester group (Figure 8f).

Doping with new components is considered to be another effective method to help form high-quality and stable perovskite films, so some researchers have adopted doping strategies to reduce the LSE. Wu et al. first demonstrated a route to suppress the LSE in PSCs through active layer engineering by incorporating slightly overstochiometric PbI_2 in precursor to tune the concentration of halide vacancies in $CsPb(I_{1-x}Br_x)_3$, which resulted in a champion efficiency of 18.14% (V_{oc} = 1.22 V, FF = 82.07%, and J_{sc} = 18.16 mA cm^{-2}) and significantly reduced LS time.[50] As shown in Figure 8g−j, with the increase of excessive PbI_2 concentration, the average grain size increases noticeably, contributing to reduced defect density and the lessened LSE. Zheng et al. confirmed that the K^+-doped ($Cs_{0.05}(FA_{0.85}MA_{0.15})_{0.95}Pb(I_{0.85}Br_{0.15})_3$) perovskites exhibit improved PCE and negligible LSE.[94] K^+ doping of perovskite films plays an essential role in reducing the trapping defect density and confining the ion migration. Li et al. fabricated PSCs doped with 5% Cs^+ to achieve an increased maximum PCE of 22.4% and inhibit the LSE under continuous light illumination.[43] Due to the addition of Cs^+, the density of accumulated mobile ions at the interfaces and nonradiative recombination defects are decreased, a consequence being that the LSE in FAMA PSCs devices is suppressed. Herterich et al. suggested that ionic liquid additive 1-butyl-3-methylimidazolium tetrafluoroborate ($BMIMBF_4$) can passivate the perovskite bulk and slow down the LS behavior via hindering ion movement.[33]

Considering high defects on the perovskite surface, surface passivation of perovskite is an effective strategy to reduce the LSE of PSCs. Shao et al. proved that improving the perovskite film morphology which has high compactness is an efficient way to reduce the LSE because grains are fused and trap states are reduced in compact films.[76] Noel et al. achieved both higher efficiency and less pronounced LSE by treating the surface of crystals with Lewis bases thiophene and pyridine compared to the control device.[103] Through Lewis base passivation, the structural defects are passivated and the nonradiative electron−hole recombination is reduced significantly in $MAPbI_{3-x}Cl_x$ films due to the electronic passivation of low-coordinated Pb atoms in the crystal.

3.3. HTL. Recent work demonstrates that the categories of HTL materials and their fabrication processes could affect the device performance and stability.[104−106] For example, Mamun et al. observed severe LSE in NiO-based PSCs because of the defects at the NiO/perovskite interface that trap charges. They presented a convenient and low-cost process for fabricating NiO films as the HTL.[107] This method of preparing NiO precursor solution by a nontoxic way was achieved by simply mixing NiO powder and HCl in an atmospheric environment. The high defects in NiO prepared from this conventional method cause significant LSE in devices. In contrast, negligible LSE was obtained in the devices with high-quality NiO films from hot-casting technique. They concluded that the NiO films with low defects and high uniformity showed a small LSE when a hot-casting temperature was below 100 °C. In addition, the strong Ni−O octahedral bonding promotes the chemical ordering of films, which facilitates charge extraction at the NiO/perovskite interface, resulting in further reduced LSE.

4. CONCLUSION AND OUTLOOK

Up to now, PSCs have become one of the most promising technologies in the photovoltaic filed due to their superior photoelectric performance and low fabrication cost. Remarkable PCEs approaching 26% have been achieved during the past decade. However, their instability in the exposure of the external environment has been a big challenge for their commercial application. In particular, the LSEs observed in a large number of solar cells have aroused heated discussions in recent years. The LSE manifests as the permanent or reversible performance improvement of solar cells under a few seconds or minutes of light illumination. Hence, it is of significant importance to study the internal mechanism of the LSE and summarize the corresponding elimination methods.

In this review, we provide a discussion of the underlying mechanisms of light soaking occurring in PSCs, which can be broadly classified as follows: (a) light can induce ions migration (including cations and halide ions); (b) light exposure can reduce the trap density at the interface, grain boundary, and bulk of perovskite through trap filling or trap deactivation process; (c) light can induce lattice expansion, thus lowering the energy barrier and reducing the nonradiative recombination; (d) light can reduce or increase the charge carriers accumulation. More than that, we also summarize the representative measures such as interfacial modification, doping, and passivation strategies to provide a further understanding to eliminate the LSE and achieve better efficiency. Overall, the generation of LSE in solar cell devices is mostly related to the change of status of the mobile charge carriers, either ions or electrons, which could exist in the bulk of functional layers and/or at their interfaces. Therefore, the mechanism of LSE could vary from case to case, and accordingly different strategies would be required to eliminate the LSE.

Although LSE is a phenomenon of PCE increase with prolonged illumination time, it significantly limits the reliability of device output power. Different mechanisms and strategies have been developed for understanding and mitigating the LSE, but the complex nature of polycrystalline perovskite and their interfaces with contact materials induces significant stability risks of devices under practical operation. More attention should be paid to screening the ions migration and to reducing the defects at the perovskite surface and interfaces. First, in the perovskite active layer, the development of phase-pure perovskite crystals (for example $FAPbI_3$) without mixing hybrid halides or cations is beneficial for obtaining LSE-free devices. For example, fine optimization of the perovskite crystallization process to generate thin film with high compactness and grain uniformity would favor enhancing the charge collection in the devices. Perovskite surface passivation by interlayer is also critical to reduce the defects concentration at interfaces in order to avoid carrier accumulation or recombination. Second, dopant-free charge transport layers with high electrical conductivity are highly preferred for mitigating the light-soaking behavior by suppression of ionic migration through these layers. Furthermore, a simplified device architecture without charge transport layers would be another way to regulate the light-soaking response of PSCs. Third, the interface between metal electrode with charge transport layer is another potential threat to limit the photostability of devices, given that air-involved doping of Spiro-OMeTAD was often occurred in n−i−p PSCs. And light-induced metal diffusion could lead to variation of contact resistance between their interfaces. Therefore, the insertion of interface or development of air-stable metal contact is of great important to yield stable and LSE-free PSCs.

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Notes

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